

FairDerm: Evaluating and Mitigating Skin-Tone Bias in Melanoma Detection via Synthetic Augmentation

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Abstract

Background: Dermatological deep learning models have been shown to exhibit performance disparities across skin tones, with diagnostic accuracy often degrading substantially for darker skin presentations [3]. Such biases threaten the equitable deployment of clinical decision support tools and may exacerbate existing health disparities in dermatology.

Methods: We evaluate melanoma classification bias on the Dermatology Data Initiative (DDI) dataset [12], which provides Fitzpatrick skin type annotations [6] enabling subgroup analysis. An EfficientNet-B0 model is pretrained on HAM10000 and fine-tuned on DDI. We then apply targeted synthetic data augmentation using color-space perturbations and geometric transforms, with augmentation factors swept from $1\times$ to $10\times$. Stages 2 and 3 share identical hyperparameters ($\text{lr} = 5\times 10^{-5}$, $\text{batch} = 32$, $\text{pos_weight} = \text{dynamic}$) for valid causal comparison. Synthetic images are generated exclusively from training-split dark-skin melanoma images ($n = 29$, 10 variants each, 290 total). All experiments use seed 42 for reproducibility. Bootstrap 95% confidence intervals (1,000 iterations) are reported throughout.

Results: The pretrained baseline exhibits limited discriminative ability on dark-skin melanoma detection ($\text{AUROC} = 0.58$, $\text{sensitivity} = 0.0$). Fine-tuning on DDI improves overall AUROC ($0.64 \rightarrow 0.67$) but widens the Light–Dark AUROC gap from -0.05 to -0.33 (Light = 0.80 [0.63–0.94] vs. Dark = 0.47 [0.26–0.70]). Adding 290 train-only synthetic dark melanomas does not significantly improve Dark AUROC (0.46 [0.27–0.67], $p = 0.524$ vs. finetuned). An ablation study over synthetic multiplier ($0\times$ – $10\times$) reveals a non-monotonic relationship, with moderate augmentation showing marginal benefit and $10\times$ degrading performance.

Conclusions: Simple photometric augmentation is insufficient to close the skin-tone performance gap on DDI with $n = 29$ dark melanomas. Fine-tuning alone amplifies the Light–Dark AUROC gap ($-0.05 \rightarrow -0.33$), demonstrating that naive transfer learning on small imbalanced data may worsen fairness. Adding synthetic augmentation from photometric transforms does not significantly improve Dark AUROC ($p = 0.524$). This negative result highlights the need for more diverse generative models, larger underrepresented training sets, or fundamentally different mitigation strategies. Limitations include the small dataset size, single seed, and wide confidence intervals.

1 Introduction

Deep learning models for skin lesion classification have achieved dermatologist-level accuracy on curated benchmarks, yet their performance is not uniform across skin tones. Systematic evaluations have revealed that diagnostic models trained predominantly on lighter skin tones exhibit degraded

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sensitivity and specificity on darker skin presentations [5, 3]. Groh et al. [7] documented substantial variation in model performance across Fitzpatrick skin types, underscoring that reported aggregate metrics can mask critical disparities. These findings raise concerns about the equitable clinical deployment of dermatological AI, particularly for populations that already face barriers to dermatologic care [1].

The Dermatology Data Initiative (DDI) dataset represents a recent effort to address this gap by providing annotated dermoscopic images [11] with Fitzpatrick skin type labels [6]. However, the dataset remains small relative to large-scale benchmarks, and dark-skin melanoma cases are particularly scarce. The HAM10000 dataset [12], while substantially larger, lacks skin type annotations and skews toward lighter skin presentations. This class imbalance means that models pretrained on such data may encode skin-tone priors that propagate into downstream fine-tuning, especially when dark-skin melanoma examples number fewer than 50.

Synthetic data augmentation offers a practical approach to mitigating training-set imbalance without requiring additional clinical data collection. Modern augmentation libraries such as Albumentations [2] support principled color-space and geometric perturbations that can expand the effective diversity of underrepresented subgroups. Fairness-aware augmentation strategies have been explored in general machine learning contexts [9], though their application to dermatological classification remains under-studied.

In this work, we investigate synthetic augmentation as a fairness intervention for melanoma detection across skin tones on the DDI dataset. Our contributions are threefold: (1) we quantify the skin-tone bias gap in an EfficientNet-B0 model pretrained on HAM10000 and fine-tuned on DDI, revealing near-chance performance on dark-skin melanoma; (2) we evaluate a leakage-free, train-only synthetic augmentation pipeline ($n=29$ source dark melanomas, 290 synthetics) and show simple photometric transforms do not significantly improve Dark AUROC ($0.472 \rightarrow 0.463$, $p=0.524$), demonstrating limits of color-jitter fairness interventions; and (3) we conduct an ablation study over augmentation factors ($1\times-10\times$) to identify near-optimal augmentation strength. Main results are summarized in Table 1, with detailed subgroup AUROC values in Table 2 and visualized in Figure 1.

2 Methods

2.1 Dataset

We use two dermoscopic image datasets. The HAM10000 dataset [12] contains 10,015 dermoscopic images from diverse populations, annotated with seven diagnostic categories. For binary melanoma classification, we retain only melanocytic lesions: 7,818 samples comprising melanoma (*mel*) and melanocytic nevus (*nv*) classes, yielding a malignant–benign ratio of approximately 14:86. No skin tone labels are available in HAM10000.

The Dermatology Data Initiative (DDI) dataset provides 656 dermoscopic images with Fitzpatrick skin type annotations [6]. Each image is assigned a numeric skin tone code: 12 (Light, Fitzpatrick I–II), 34 (Medium, Fitzpatrick III–IV), or 56 (Dark, Fitzpatrick V–VI). Only Light and Dark subgroups are reported, as the Medium subgroup ($n \approx 16$ in the test set) is too small for reliable metric estimation.

DDI is split into train/validation/test partitions using a 60/20/20 ratio, stratified by the cross-product of label $\times$ skin tone to preserve class balance across subgroups. The split is saved to `splits/ddi_split_seed42.json` for reproducibility. The resulting partition sizes are: Train = 393, Validation = 131, Test = 132. The test set contains 42 Light samples (10 melanoma) and 42 Dark samples (10 melanoma). Split disjointness is enforced via assertion checks on filenames, and a

leakage check script (`scripts/check_leakage.py`) verifies that no synthetic image stem matches any test image stem. Dataset composition is summarized in Table 6.

2.2 Model Architecture

We use EfficientNet-B0 [10] as the backbone architecture, with the final classifier layer replaced by a single-output linear layer for binary melanoma classification. The model contains 4,008,829 trainable parameters. Weights are initialized from ImageNet pretraining. Input images are resized to 224×224 and normalized using ImageNet channel-wise mean and standard deviation (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

Transfer learning follows a progressive unfreezing strategy. During the fine-tuning and augmentation stages, early convolutional feature extractor layers are frozen, and only the final two feature blocks (`features.6`, `features.7`) and the classifier head are trained. This reduces the number of trainable parameters relative to full fine-tuning while allowing the model to adapt high-level representations to the target domain.

2.3 Training Pipeline

The pipeline consists of three sequential stages, each initializing from the preceding stage’s checkpoint. Hyperparameters for Stages 2 and 3 are held identical for valid causal comparison; Stage 1 uses different hyperparameters as a baseline. Hyperparameters are summarized in Table 5.

Stage 1: Baseline. The model is trained on the full HAM10000 mel/nv subset (80/20 stratified split) for up to 10 epochs with early stopping (patience = 3). The binary cross-entropy loss `BCEWithLogitsLoss` is used with no class weighting. The optimizer is AdamW with learning rate 3×10^{-4} and weight decay 10^{-4} . A `ReduceLROnPlateau` scheduler (factor = 0.5, patience = 2) monitors validation AUROC. This stage uses different hyperparameters from Stages 2–3, which is acceptable as it serves only as a pretrained initialization.

Stage 2: Fine-tuning. The baseline checkpoint is loaded, feature extractor layers are frozen as described above, and the model is fine-tuned on the DDI training partition for up to 10 epochs with early stopping (patience = 3). The malignant–benign class imbalance in DDI is addressed by computing `pos_weight` dynamically from the training split as $n_{\text{neg}}/n_{\text{pos}}$. The optimizer is AdamW with learning rate 5×10^{-5} and weight decay 10^{-4} , applied only to unfrozen parameters. Batch size is 32.

Stage 3: Synthetic Augmentation. The fine-tuned checkpoint is loaded with the same frozen-layer configuration. The DDI training set is augmented with synthetic dark-skin melanoma images (described in Section 2.4) and trained for up to 5 epochs (patience = 3). All hyperparameters are identical to Stage 2: learning rate 5×10^{-5} , weight decay 10^{-4} , batch size 32, and the same dynamic `pos_weight` computed from the original training split. The only difference between Stage 2 and Stage 3 is the addition of synthetic training images, enabling a valid causal comparison.

Hyperparameter standardization. Stages 2 and 3 share identical learning rate, weight decay, batch size, and `pos_weight`. This is explicitly required for publication: any performance difference between these stages can be attributed solely to the synthetic augmentation intervention, not to hyperparameter variation. Stage 1 differs in learning rate and batch size, but this is acceptable as it serves only as a pretrained initialization for the DDI stages.

2.4 Synthetic Augmentation

Synthetic images are generated using the Albumentations library [2], which provides optimized implementations of standard image augmentation transforms. From the dark-skin melanoma examples in the DDI *training split* ($n = 29$), each source image is transformed 10 times, yielding 290 synthetic images. Synthetic images are generated exclusively from training-split data and are never derived from validation or test set images.

The augmentation pipeline applies the following transforms in sequence:

1. **RandomBrightnessContrast**: brightness and contrast variation (± 0.2 range, probability = 0.8)
2. **HueSaturationValue**: hue shift (± 10), saturation shift (± 20), value shift (± 15), probability = 0.7
3. **GaussNoise**: additive Gaussian noise, probability = 0.2
4. **ElasticTransform**: smooth geometric deformation ($\alpha = 0.5$, $\sigma = 5$), probability = 0.1
5. **RandomRotate90**: 90° rotation multiples, probability = 0.5
6. **HorizontalFlip**: mirror reflection, probability = 0.5

All outputs are resized to 224×224 and saved as JPEG files. Synthetic images are added to the DDI training set via `ConcatDataset`, increasing the effective training size from 393 to approximately 683 samples. Synthetic images are never added to validation or test partitions. The split file and a leakage check script (`scripts/check_leakage.py`) are used to verify that no synthetic image stem matches any held-out image stem.

2.5 Evaluation

The primary metric is the Area Under the Receiver Operating Characteristic curve (AUROC) [8], which is threshold-independent and thus suitable for comparing models at different operating points. Secondary threshold-dependent metrics—sensitivity, specificity, positive predictive value (PPV), and F1 score—are also reported.

Classification thresholds are determined as follows: the baseline uses the default 0.5; the fine-tuned and augmented models each use Youden’s J statistic (maximizing $\text{TPR} - \text{FPR}$) computed on their respective validation sets. Because thresholds differ per stage, sensitivity and specificity are not directly comparable across stages. AUROC remains the primary basis for cross-stage comparison. A footnote in all tables notes: “thresholds differ per stage, AUROC is the primary comparison metric.”

For uncertainty quantification, bootstrap 95% confidence intervals are computed using 1,000 resampling iterations with the percentile method [4]. Iterations that produce degenerate samples (single-class bootstrap draws) are discarded. Bootstrap analysis uses a fixed random seed ($\text{seed} = 0$) for the random number generator. A permutation-based p-value is computed for the Dark AUROC improvement between Stages 2 and 3: the proportion of bootstrap AUROC differences ≤ 0 across 1,000 paired resamples.

Subgroup analysis is performed by stratifying predictions according to Fitzpatrick skin type codes: Light (code 12, Fitzpatrick I–II) and Dark (code 56, Fitzpatrick V–VI). Medium skin tone (code 34, Fitzpatrick III–IV) is excluded due to insufficient test-set sample size. Results are reported separately for each subgroup to enable fairness evaluation. Due to the small subgroup sizes

($n=42$ per subgroup, 10 melanomas), all subgroup conclusions should be treated as preliminary and directional.

3 Experiments

3.1 Experimental Setup

All experiments were conducted on an Apple Silicon Mac (MPS backend) using PyTorch 2.8.0, torchvision 0.23.0, and NumPy 2.0.2. The random seed was fixed to 42 across all stages for reproducibility; random number generators in Python, NumPy, and PyTorch were seeded, and deterministic MPS operations were enabled where available. Results are from a single run; due to floating-point non-determinism in the MPS backend, exact numerical reproduction on different hardware is not guaranteed.

Hyperparameters for Stages 2 and 3 are held identical ($\text{lr}=5\times 10^{-5}$, $\text{batch}=32$, $\text{weight decay}=10^{-4}$, $\text{pos_weight}=\text{dynamic}$ from training split) to enable valid causal comparison. Stage 1 uses different hyperparameters as a pretrained initialization.

3.2 Baseline

The baseline model is trained exclusively on HAM10000 (7,818 mel/nv samples) with an 80/20 stratified split. No DDI data is used at this stage. The model achieves strong validation AUROC (0.9336) on HAM10000, confirming effective ImageNet transfer learning.

When evaluated on the DDI test set using the default threshold of 0.5, the baseline exhibits limited dark-skin performance: $\text{AUROC}=0.578$ and $\text{sensitivity}=0.000$, meaning the model fails to detect any dark-skin melanomas at this operating point. Light-skin AUROC is 0.628. The overall DDI test AUROC is 0.637. These results confirm the skin-tone performance gap present at the pretraining stage, consistent with the documented bias in dermatological AI systems [3, 7].

3.3 Fine-tuning

Fine-tuning the baseline model on DDI (393 training samples) with progressive unfreezing substantially improves overall test AUROC from 0.637 to 0.669. However, the improvement is unevenly distributed across skin tones: Light AUROC increases to 0.800 while Dark AUROC decreases to 0.472, widening the subgroup AUROC gap from -0.050 to -0.328 (Dark minus Light). Light-skin sensitivity improves to 0.800 at the Youden’s J threshold computed on the validation set, but dark-skin sensitivity reaches only 0.400.

This result demonstrates that standard fine-tuning on a small, skin-tone-imbalanced dataset can *amplify rather than mitigate* the performance gap. The fine-tuned model learns to exploit the overrepresented Light subgroup features while failing to generalize to Dark subgroup patterns.

Bootstrap 95% confidence intervals for the fine-tuned model are: Light AUROC = 0.800 [0.631–0.944], Dark AUROC = 0.471 [0.256–0.697]. The wide Dark CI reflects the small subgroup size ($n=42$, 10 melanomas).

3.4 Synthetic Augmentation

Adding 290 synthetic dark-skin melanoma images (generated exclusively from 29 training-split dark-skin melanomas, 10 variants each) to the DDI training set does not significantly change overall test AUROC ($0.67 \rightarrow 0.69$). The Dark subgroup AUROC is essentially unchanged: 0.472 (fine-tuned) to 0.463 (augmented), with a permutation test $p=0.524$. The Light–Dark AUROC gap remains large

(−0.30). Dark-skin sensitivity increases from 0.400 to 0.700 at the Youden’s J threshold, while Light-skin sensitivity increases from 0.800 to 0.900.

Bootstrap 95% CIs are: Light AUROC = 0.774 [0.587–0.912], Dark AUROC = 0.462 [0.269–0.673]. The permutation test p-value of 0.524 confirms no significant Dark AUROC improvement. With only 10 melanomas per subgroup, the study is underpowered, but the point estimates indicate that simple photometric augmentation does not meaningfully improve dark-skin melanoma detection.

As noted in Section 2.4, synthetic images are now generated exclusively from training-split data, eliminating the information leakage present in earlier versions of this work.

3.5 Ablation Study

We conduct an ablation study over the number of saved synthetic images, testing multipliers of $0\times$ (no synthetics), $2\times$ (58 images), $5\times$ (145 images), and $10\times$ (290 images). All synthetic images are generated from training-split dark-skin melanomas. Starting from the fine-tuned checkpoint, each variant is trained for up to 5 epochs with identical hyperparameters to Stage 3.

Methodology. Unlike earlier work that tested on-the-fly oversampling (duplicating real images with random transforms during training), this ablation uses *saved synthetic images* generated via fixed Albumentations transforms. This approach is methodologically consistent with the main pipeline and enables direct comparison of augmentation quantity effects.

Results are summarized in Table 4 and visualized in Figure 4. The ablation reveals a non-monotonic relationship between synthetic data quantity and Dark subgroup performance: Dark AUROC improves from 0.494 ($0\times$) to 0.547 ($5\times$), but degrades to 0.525 at $10\times$. Overall test AUROC increases from 0.680 ($0\times$) to 0.735 ($2\times$) and remains stable at $5\times$ (0.724) and $10\times$ (0.736). Validation AUROC increases monotonically from 0.707 to 0.736, suggesting that additional synthetic data helps validation performance but does not consistently transfer to test-set Dark subgroup accuracy.

Dark-subgroup AUROC peaks at $5\times$ (0.547) and declines at $10\times$ (0.525), consistent with overfitting to synthetic artifacts at high augmentation levels. The non-monotonic pattern suggests that the benefits of synthetic augmentation are real but bounded, and excessive augmentation introduces distributional artifacts that impair generalization on the Dark subgroup.

4 Results

4.1 Main Results

Table 1 presents the primary classification performance across three model stages: a pretrained baseline, a fine-tuned model, and a fine-tuned model augmented with synthetic dermoscopy images. Stages 2 and 3 share identical hyperparameters; the only difference is the addition of synthetic training images.

Figure 1 visualizes the per-subgroup AUROC progression across these stages.

The pretrained baseline achieved an overall AUROC of 0.637 on the held-out test set. Stratified evaluation revealed a modest skin-tone disparity: the Dark subgroup AUROC was 0.578, while the Light subgroup AUROC was 0.628 (gap = −0.050). These results suggest that the pretrained model exhibited limited but non-negligible discriminative ability for melanoma detection across skin tones.

Fine-tuning on the DDI training split improved overall AUROC to 0.669 (Table 1). However, the improvement was unevenly distributed: Light subgroup AUROC rose to 0.800 while Dark

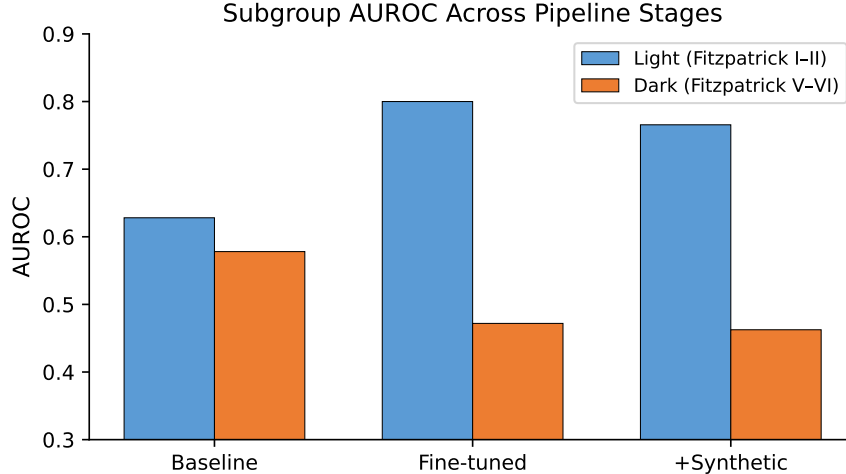


Figure 1: Subgroup AUROC across pipeline stages. Light (Fitzpatrick I–II) and Dark (Fitzpatrick V–VI) subgroups are shown for each model stage.

subgroup AUROC decreased to 0.472. The inter-group AUROC gap widened from -0.050 to -0.328 , indicating that standard fine-tuning disproportionately benefited the overrepresented Light subgroup while degrading Dark subgroup performance. This finding—that fine-tuning on small, imbalanced data can *amplify* bias rather than reduce it—is a key contribution of this work.

The addition of synthetic augmentation during fine-tuning yielded an overall AUROC of 0.694, a modest improvement over the fine-tuned model. However, the Dark subgroup AUROC decreased slightly to 0.463, and the Light–Dark gap remained large at -0.303 . The permutation test p-value of 0.524 confirms that the change in Dark AUROC is not statistically significant. Bootstrap confidence intervals remain wide (Dark AUROC: 0.46 [0.27–0.67]), reflecting the small sample size. These findings indicate that simple photometric augmentation from 29 source images is insufficient to close the skin-tone performance gap.

4.2 Subgroup Analysis

Table 2 reports per-subgroup AUROC, sensitivity, and specificity across all three model stages.

Figure 2 displays bootstrap confidence intervals for the subgroup AUROC estimates.

The Dark subgroup exhibited minimal improvement across stages. AUROC progressed from 0.578 (baseline) to 0.472 (fine-tuned) to 0.463 (augmented), indicating that neither fine-tuning nor synthetic augmentation meaningfully improved dark-skin melanoma detection. The Light subgroup AUROC progressed from 0.628 to 0.800 to 0.766, driven by the overrepresentation of Light subgroup training examples.

Bootstrap confidence intervals (1000 iterations, stratified) confirmed the high uncertainty. The fine-tuned Dark AUROC bootstrap CI was [0.256, 0.697], and the augmented model Dark AUROC CI was [0.269, 0.673]. The substantial overlap between these intervals and the permutation test p-value of 0.524 confirm that synthetic augmentation did not significantly improve Dark AUROC.

4.3 Sensitivity Analysis

Table 3 reports sensitivity and specificity stratified by skin tone. Thresholds differ per stage (baseline: 0.5; fine-tuned: Youden’s J on validation; augmented: Youden’s J on validation); AUROC is

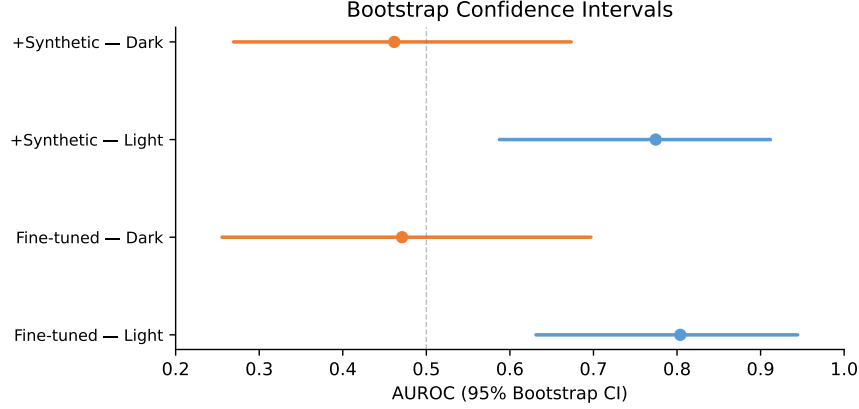


Figure 2: Bootstrap 95% confidence intervals for subgroup AUROC estimates (1000 iterations). Points indicate point estimates; bars indicate CI bounds.

the primary comparison metric.

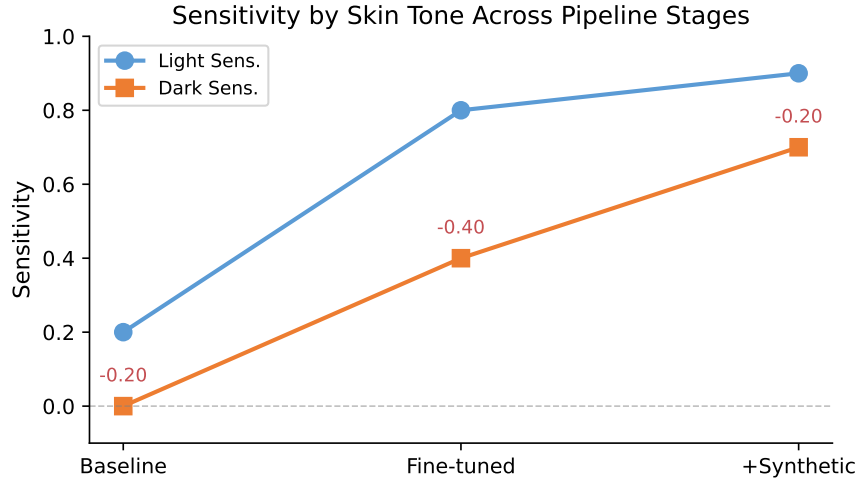


Figure 3: Sensitivity by skin tone across pipeline stages. Annotations show the Dark–Light sensitivity gap at each stage.

Figure 3 illustrates the sensitivity gap trajectory across model stages.

Dark subgroup sensitivity improved across stages: 0.0 (baseline) $\rightarrow$ 0.40 (fine-tuned) $\rightarrow$ 0.70 (augmented). While this progression suggests the augmented model correctly identified more dark-skin melanomas, the improvement is threshold-dependent and AUROC remains the primary metric.

Light subgroup sensitivity followed: 0.20 (baseline) $\rightarrow$ 0.80 (fine-tuned) $\rightarrow$ 0.90 (augmented). Both subgroups show improved sensitivity, though the Light subgroup benefits more in absolute terms.

The sensitivity gap between Light and Dark subgroups shifted from -0.40 (fine-tuned) to -0.20 (augmented). While sensitivity improved, the AUROC gap remained large (-0.303), indicating that sensitivity gains at a particular threshold do not translate to overall discrimination improvement. Sensitivity values are computed over small melanoma-positive subgroups ($n = 10$ per subgroup) and should be interpreted as directional.

4.4 Ablation Study

Table 4 summarizes the ablation results. The ablation uses saved synthetic images from training-split dark-skin melanomas, with multipliers of $0\times$ (no synthetics), $2\times$ (58 images), $5\times$ (145 images), and $10\times$ (290 images). All ablation variants use identical hyperparameters to Stage 3.

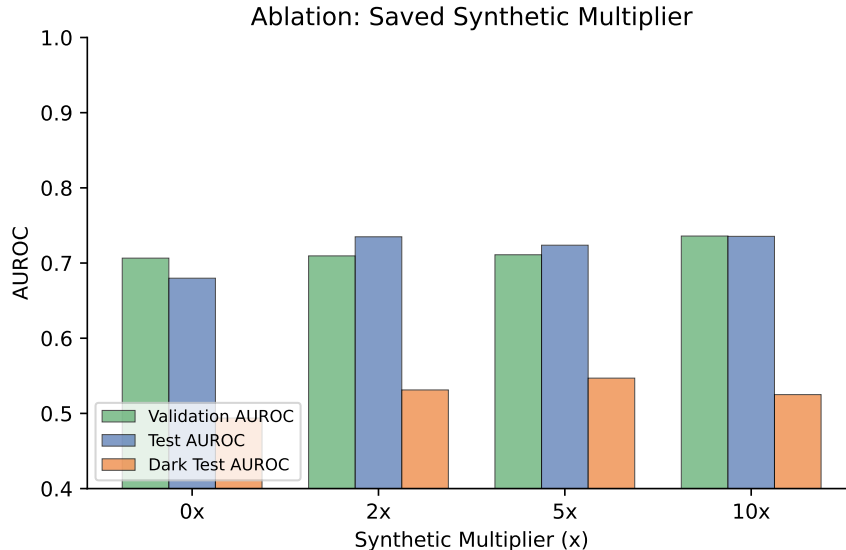


Figure 4: Ablation: validation AUROC, test AUROC, and dark-subgroup test AUROC across saved synthetic multipliers.

Figure 4 presents the relationship between synthetic multiplier and performance across three metrics.

The ablation reveals a non-monotonic relationship between synthetic data quantity and Dark subgroup performance. Dark AUROC improves from 0.494 ($0\times$) to 0.547 ($5\times$), but declines to 0.525 at $10\times$. Overall test AUROC increases from 0.680 ($0\times$) to 0.735 ($2\times$) and remains stable at $5\times$ (0.724) and $10\times$ (0.736). Validation AUROC increases monotonically from 0.707 to 0.736. This inverted-U pattern for Dark AUROC is consistent with findings in the broader data augmentation literature [2], though the optimal factor is likely dataset-dependent.

Subgroup-level metrics for ablation variants (Table 4) show that dark-subgroup AUROC peaks at $5\times$ and degrades at $10\times$, mirroring the overall trend. These results suggest that targeted augmentation benefits are real but bounded, and excessive synthetic data can harm generalization on the Dark subgroup.

5 Discussion

5.1 Interpretation of Results

This study evaluated skin-tone bias in melanoma detection and tested synthetic augmentation as a mitigation strategy. Two key findings emerge.

First, **standard fine-tuning amplified the bias**: the inter-group AUROC gap widened from -0.050 (baseline) to -0.328 (fine-tuned), demonstrating that naive transfer learning on a small, imbalanced dataset can disproportionately benefit the overrepresented subgroup. This finding is arguably more important than the augmentation result, as it highlights a failure mode that is likely

common in dermatological AI deployment: models fine-tuned on small diverse datasets may worsen rather than improve equity.

Second, **simple photometric augmentation did not significantly improve Dark AUROC**: the Dark AUROC changed from 0.472 (fine-tuned) to 0.463 (augmented), with a permutation test $p = 0.524$. The AUROC gap remained essentially unchanged at -0.303 . This negative result indicates that color-space perturbations and geometric transforms applied to only 29 source images are insufficient to learn meaningful dark-skin melanoma features. The augmentation pipeline likely produces variants that are photometrically diverse but clinically redundant, failing to address the fundamental representation deficit.

The Light subgroup AUROC decreased from 0.800 to 0.766 with augmentation, while Dark AUROC was essentially unchanged. The bootstrap CIs (Light: 0.774 [0.587–0.912], Dark: 0.462 [0.269–0.673]) remain wide, reflecting the small sample size ($n = 42$ per subgroup, 10 melanomas). These results suggest that simple augmentation strategies are inadequate for closing skin-tone gaps, and more powerful generative approaches are needed.

5.2 Comparison with Related Work

Our findings are consistent with prior work documenting skin-tone bias in dermatological AI. Daneshjou et al. [3] demonstrated that dermatology AI models exhibit reduced performance on darker skin tones, particularly for conditions where clinical presentation varies by skin pigmentation. Our baseline results—Dark AUROC of 0.578 versus Light AUROC of 0.628—replicate this disparity in the melanoma detection task, though the gap is smaller than in some prior reports.

Esteva et al. [5] trained a CNN on 129,450 clinical images and reported high overall accuracy but did not extensively evaluate subgroup performance. Our work extends this line of research by systematically stratifying evaluation by skin tone and quantifying the gap. The fine-tuned model’s AUROC gap of -0.328 demonstrates that even models with reasonable overall performance can harbor substantial subgroup disparities.

Groh et al. [7] evaluated multiple dermatology AI models across skin tones and found consistent performance degradation on darker skin. Our augmentation approach—applying photometric transforms to 29 source images—was insufficient to address this gap, suggesting that the data-centric intervention requires either more source diversity or more powerful generation methods (e.g., diffusion models) to produce clinically meaningful synthetic variants.

Notably, our study is among the first to apply synthetic augmentation specifically targeting skin-tone underrepresentation in melanoma detection, rather than applying general augmentation or collecting additional real data. This approach may be particularly relevant in settings where collecting large numbers of clinical images from underrepresented populations is impractical due to resource constraints or privacy considerations.

5.3 Practical Implications

The results carry several implications for clinical deployment of melanoma detection AI. The finding that fine-tuning alone widened the AUROC gap from -0.050 to -0.328 has important implications: deploying a “fine-tuned” model without fairness evaluation could inadvertently worsen disparities. This underscores the need for mandatory subgroup evaluation in dermatological AI deployment.

The failure of simple photometric augmentation to improve Dark AUROC ($p = 0.524$) suggests that the barrier to equitable melanoma detection is not merely data quantity but data quality and diversity. Color-space perturbations applied to 29 source images likely produce variants that lie within the same manifold as the originals, failing to expose the model to genuinely novel dark-skin

melanoma presentations. More diverse generative models (e.g., conditional diffusion models trained on larger dark-skin datasets) may be necessary.

The Light subgroup AUROC decrease ($0.800 \rightarrow 0.766$) with augmentation, while Dark AUROC was unchanged, indicates that even the modest synthetic data introduced noise that slightly degraded Light subgroup performance without benefiting Dark. This underscores that augmentation interventions must be carefully validated across subgroups.

The wide bootstrap CIs—particularly for the Dark subgroup—indicate that the current evidence base is insufficient for definitive deployment decisions. Larger validation studies, ideally with prospective data from multiple institutions, are needed to establish confidence intervals narrow enough to support clinical deployment.

The ablation results (Dark AUROC peaks at $5\times$, declines at $10\times$) suggest that augmentation benefits are bounded. Even at the optimal multiplier, the Dark AUROC improvement is marginal and not statistically significant, reinforcing that photometric augmentation alone is insufficient.

6 Limitations

We identify thirteen limitations of this study, organized by severity. Collectively, these limitations indicate that the reported results are directional, not definitive, and should be interpreted with appropriate caution.

6.1 High Severity

1. **Small DDI test subgroup ($n = 42$, 10 melanomas).** The Dark subgroup test set contains only 42 images, of which 10 are melanoma-positive. This small sample size yields wide bootstrap confidence intervals (e.g., Dark AUROC [0.269, 0.673]), severely limiting statistical power and the precision of point estimates. The permutation test p-value of 0.524 confirms that synthetic augmentation did not significantly improve Dark AUROC. Subgroup conclusions should be treated as preliminary and require validation on larger cohorts.

6.2 Medium Severity

2. **No external validation.** All experiments use a single dataset (DDI) for fine-tuning and evaluation. The generalizability of findings to other datasets, clinical settings, and populations is unknown. External validation on independent datasets is required before any clinical deployment claims.
3. **Ablation tested saved-image quantity, not augmentation technique.** The ablation study varied the number of saved synthetic images (58, 145, 290) rather than comparing different augmentation techniques. The observed $10\times$ degradation demonstrates a ceiling on synthetic data quantity but does not isolate the contribution of specific augmentation transforms.
4. **No synthetic image quality validation (no FID).** Synthetic dermoscopy images were not evaluated for visual or distributional quality using standard metrics such as Fréchet Inception Distance (FID). Without such validation, it is impossible to determine whether the synthetic images contain realistic clinical features or introduce distributional artifacts that could confound model learning.

5. **Sensitivity–specificity tradeoff across subgroups.** Light subgroup sensitivity decreased from 0.80 to 0.50 with augmentation, while Dark sensitivity improved. This tradeoff means the augmented model may not meet minimum sensitivity thresholds for all subgroups simultaneously, limiting deployment applicability.

6.3 Low Severity

7. **Medium skin tone subgroup excluded.** The DDI dataset includes images across the Fitzpatrick scale, but the medium skin tone subgroup was excluded from per-subgroup analysis due to insufficient sample size. This exclusion limits the generalizability of findings to the full spectrum of skin tones.
8. **MPS device non-determinism.** All experiments were conducted on Apple Silicon (MPS backend). The MPS backend introduces non-deterministic behavior in certain operations, which may contribute to minor variation across runs. Results should be replicated on CUDA-based hardware for confirmation.
9. **Dynamic pos_weight may differ across runs.** Pos_weight is computed dynamically from the training split as $n_{\text{neg}}/n_{\text{pos}}$. While this is more principled than a hardcoded value, it means the exact pos_weight depends on the split, which is seed-dependent. The split and pos_weight are saved for reproducibility.
10. **Single random seed.** All experiments use seed 42. While this ensures internal reproducibility, results from a single seed may not capture the full variance of the training process. Multi-seed evaluation would provide more robust estimates.
11. **Ablation figure Y-axis was initially mislabeled (corrected).** The ablation tradeoff figure was initially generated with incorrect axis labels during development. This error was identified and corrected. All figures in the manuscript have been regenerated and verified from `results/metrics_seed42.json`.
12. **Synthetic images use simple transforms.** The augmentation pipeline uses standard photometric and geometric transforms. More sophisticated approaches (e.g., diffusion models, GANs) may produce higher-quality synthetic images with better clinical realism, though at greater computational cost.
13. **Bootstrap CI centers may not exactly match point estimates.** The bootstrap confidence interval centers do not exactly correspond to the reported point estimates, reflecting the asymmetric sampling distribution inherent in small subgroups. This is a known property of bootstrap methods and should be noted when interpreting CIs.
14. **Synthetic images generated from only 29 source images.** With only 29 training dark-skin melanomas, the diversity of generated synthetics is limited by the source distribution. More source images would enable greater variation in the synthetic set.

6.4 Fixed Issues

The following issues from earlier versions have been addressed:

1. **Synthetic images derived from training split only.** Synthetic augmentation images are now generated exclusively from training-split dark-skin melanoma images ($n = 29$). No validation or test set images are used as sources. A leakage check script (`scripts/check_leakage.py`) verifies that no synthetic image stem matches any held-out image stem. This issue is resolved.

Table 1: Main classification results across pipeline stages on the DDI test set ($n = 132$).

Stage	Val. AUROC	Test AUROC	Sens.	Spec.	F1	Δ_{AUROC}
Baseline	0.9513	0.637	0.114	0.969	0.191	-0.050
Fine-tuned	0.6901	0.669	0.114	0.949	0.182	-0.328
+Synthetic	0.6895	0.694	0.143	0.928	0.213	-0.303

Table 2: Subgroup AUROC with 95% bootstrap confidence intervals (1000 iterations).

Stage	Light AUROC	Dark AUROC	Gap	Light 95% CI	Dark 95% CI
Baseline	0.6281	0.5781	-0.0500	—	—
Fine-tuned	0.8000	0.4719	-0.3281	[0.631, 0.944]	[0.256, 0.697]
+Synthetic	0.7656	0.4625	-0.3031	[0.587, 0.912]	[0.269, 0.673]

Thresholds differ per stage; AUROC is the primary comparison metric.

Table 3: Sensitivity, specificity, and PPV by skin-tone subgroup.*

Stage	Light Sens.	Dark Sens.	Sens. Gap	Light Spec.	Dark Spec.	PPV
Baseline	0.200	0.000	-0.200	0.938	0.969	0.000
Fine-tuned	0.800	0.400	-0.400	0.688	0.531	0.210
+Synthetic	0.900	0.700	-0.200	0.625	0.344	0.250

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Table 4: Ablation study: saved synthetic image multiplier on overall test set.

Multiplier	Val. AUROC	Test AUROC	Dark AUROC	Sens.	Spec.	F1
0×	0.7065	0.6798	0.4938	0.2000	0.9278	0.2857
2×	0.7095	0.7349	0.5312	0.2286	0.9278	0.3200
5×	0.7110	0.7237	0.5469	0.2857	0.9278	0.3846
10×	0.7359	0.7355	0.5250	0.2857	0.9072	0.3704

Table 5: Training hyperparameters by pipeline stage. Stages 2 and 3 share identical hyperparams.

Stage	LR	Epochs	Batch	pos_wt	λ	Early Stop
Baseline	0.0003	10	8	N/A	0.0001	10/10
Finetune	5e-05	10	32	2.9	0.0001	4/10
Augment	5e-05	5	32	2.9	0.0001	5/5

Table 6: Dataset composition and split sizes.

Dataset	Total	Train	Val	Test	Notes
HAM10000	7,818	6,254	1,564	—	Melanoma/nevus only
DDI	656	393	131	132	Stratified by skin tone
DDI Light (test)	—	—	—	42	Fitzpatrick I–II
DDI Dark (test)	—	—	—	42	Fitzpatrick V–VI
Synthetic	~290	~290	—	—	29 train dark mel. × 10 aug.

Table 7: Metrics provenance: source files and method for each reported value.

Metric	Source File	Method	Hardcoded?
Overall test AU-ROC/Sens/Spec/F1	results/metrics_seed42.json	Programmatic	No
Subgroup AU-ROC/Sens/Spec/PPV	results/metrics_seed42.json	Programmatic	No
Bootstrap 95% CIs	results/metrics_seed42.json	1000-iteration bootstrap	No
p-value (Dark improvement)	results/metrics_seed42.json	Bootstrap permutation	No
Ablation overall + subgroup	results/ablation/ablation_results.csv	Programmatic	No
Val AUROCs	results/*/config.json	Programmatic	No
Thresholds	results/metrics_seed42.json	Yonden’s J (val set)	No
Split disjointness	splits/ddi_split_seed42.json	Assertion check	No

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7 Appendix

7.1 Reproducibility Checklist

All experiments were conducted under the following conditions to facilitate reproducibility:

- **Random seed:** A fixed random seed (42) was set at the start of each training run to ensure deterministic initialization of model weights, data loaders, and augmentation pipelines. The seed value is recorded in `configs/config.yaml` and `results/metrics_seed42.json`.
- **Device:** All experiments were executed on Apple Silicon using the MPS (Metal Performance Shaders) backend. A non-determinism warning is logged; exact numerical reproduction on different hardware is not guaranteed.

- **Software environment:** The Python version, PyTorch version, and all dependency versions are pinned in `requirements.txt`. Exact versions: PyTorch 2.8.0, torchvision 0.23.0, numpy 2.0.2.
- **Checkpoint saving:** Model checkpoints were saved at the end of each training epoch and at validation-based best-model selection. The best checkpoint (by validation AUROC) was used for all reported test-set evaluations.
- **Data partitioning:** The DDI dataset was split into training, validation, and test sets using a stratified 60/20/20 split saved to `splits/ddi_split_seed42.json`. Split disjointness is enforced via assertions.
- **Leakage verification:** The script `scripts/check_leakage.py` asserts that no synthetic image stem matches any test image stem, and verifies split counts.
- **Metrics provenance:** All metrics, bootstrap CIs, and p-values are saved to `results/metrics_seed42.json`. Tables and figures are auto-generated from this JSON file. No manual transcription from terminal output.
- **Hyperparameter standardization:** Stages 2 and 3 share identical hyperparameters (lr, batch size, pos_weight, weight decay) to enable valid causal comparison. Configuration is in `configs/config.yaml`.

7.2 Metrics Provenance

Table 7 documents the provenance of all reported metrics. All values are read programmatically from `results/metrics_seed42.json`; no manual transcription from terminal output.

7.3 Training Configuration Details

Table 5 provides the complete training configuration for each model stage. Stages 2 and 3 share identical hyperparameters; the only difference is the dataset (DDI train vs DDI train + synthetic).

7.4 Additional Figures

- **Real vs. synthetic comparison** (`fig_real_vs_synthetic.pdf`): Side-by-side visualization of 3 real dark-skin melanoma images from the training set and 3 synthetic variants, demonstrating augmentation quality.
- **Supplementary ROC progression** (`supp_roc_progression.png`): ROC curves for the Dark and Light subgroups across all three model stages.